Whose Regulation Does the Model Know? Reliability of LLM-Retrieved Regulatory Information for Public Environmental Management in 25 Countries

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Abstract

A municipal environmental manager in São Paulo, Lagos, or Jakarta asks a language model what the binding $PM_{2.5}$ standard is in their country. Nothing in the answer says whether it matches the official register, and no established method exists for checking. We contribute such a protocol and report what it finds. It scores answers against official registers, adjudicating by code wherever the correct answer is a published value and by a three-vendor judge panel elsewhere. We apply it to air pollution regulation across 25 countries, 14 deployment stacks and four languages (8,300 scored cells). The three fixes an agency can deploy without building anything all fail, and the most intuitive one makes the answer worse: querying in the country's own language *lowers* accuracy by 4.8 percentage points (Wilcoxon $p = 3 \times 10^{-15}$) and makes the model 17.7 points less likely to return a register-checkable

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value at all; declaring the official role does not narrow the gap ($p = 0.27$); and a regionally tuned model ranks last of fourteen. Accuracy also collapses on the task that carries administrative consequence, retrieving the binding value: 0.37 against 0.61 for synthesis. A North/South gap persists (+5.4 pp, permutation $p = 0.020$), concentrated where answers depend on a national fact — on the national standard, Global South odds are 0.22 of Global North odds. Scope is bounded: we measure agreement with published regulation in one domain, not general capability. The protocol transfers to any regulated domain whose correct answer is published.
Keywords: algorithm auditing, geographic bias, multilingual bias, digital government, Global South

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1. Introduction

Large language models have moved from research curiosity to working infrastructure of public administration in under five years: an OECD survey of core government functions documents generative-AI adoption across policymaking and service delivery, including assistants that help public-service advisers retrieve information and the sources behind it (OECD, 2025). Air pollution policy is a domain where this carries direct stakes for public health. Fine particulate matter is among the leading environmental risk factors for premature mortality worldwide (GBD 2019 Risk Factors Collaborators, 2020; World Health Organization, 2021), and the burden falls disproportionately on the Global South. When a public environmental manager in São Paulo, Lagos, or Jakarta asks an LLM about an ambient air quality standard, an attainment deadline, or the evidence behind an intervention, an inaccurate answer is not a benign error. It can propagate into a regulatory brief, a procurement decision, or an emergency response during a pollution episode.

This paper asks whether LLMs serve Global South air pollution policy as reliably as they serve the Global North, and whether the way a query is framed changes the answer. Both premises are established. Geographic factual bias is real and measurable: error rates are 1.5× higher for Sub-Saharan African than North American countries across 20 models (Moayeri et al., 2024), model ratings on sensitive subjective topics track socioeconomic conditions up to $\rho = 0.70$ (Manvi et al., 2024), and models systematically privilege statements about the Global North (Mirza et al., 2024). And the linguistic asymmetry behind it is structural: the languages of most Global South countries sit in the under-served classes of the resource hierarchy (Joshi et al., 2020), and large Internet-derived corpora "overrepresent hegemonic viewpoints and encode biases potentially damaging to marginalized populations" (Bender et al., 2021). The asymmetry reproduces the uneven epistemic infrastructure decolonial scholarship has analysed for AI systems (Mohamed et al., 2020).

Three gaps remain open for a domain like this one, and they organise the study. No benchmark targets regulatory information with official ground truth: existing evaluations use World Bank recall, subjective ratings, journalistic fact-checking, or everyday cultural knowledge, none of which measures the tasks a manager actually delegates — stating a binding standard, retrieving an officially measured concentration, summarising health-burden evidence, identifying legal instruments and executing agencies, recommending an operational response. Persona framing is unexamined as an experimental lever, although practitioners routinely prefix a role and the vendors themselves advise it (Section 2); whether it *reduces* the gap by steering toward regulation-compliant answers or *amplifies* it by inviting confident fabrication has not been tested in a pre-specified design. And the corpus-representation mechanism is asserted rather than measured against confounders, and rarely separated into its two channels — *language-corpus* size (Xue et al., 2021; Abadji et al., 2022) against *country-coverage* breadth — a distinction that matters here because the geographic gap is measured in English.

Objectives and scope

The study has three bounded objectives. *First, to establish a protocol for auditing regulatory information*: the object is agreement between a model's answer and the value published in the official register, for tasks an environmental agency actually delegates — not model capability in general, reasoning quality, or usefulness for any other purpose. The protocol is the contribution that transfers; the domain is where we demonstrate it. *Second, to test the mitigations available to an administration*: a public body facing poor answers about its own country can realistically declare the official role in the prompt, ask in the national language, or procure a regionally tuned model, and we test all three as designed factors rather than discussing them. We do not test retrieval augmentation, fine-tuning, or human-in-the-loop verification, which require capacity most agencies in our Global South sample do not have. *Third, to locate where reliability differs and, as far as the design allows, why*: we test the corpus-representation account at the level where the measurement has resolution, and claim no causal identification.

Throughout, the unit of analysis is the model *as served* — weights, vendor, and serving infrastructure reached through a pinned endpoint — because a public manager interacts with that stack and any measured gap is a property of it. Hypotheses, sampling and analysis plan were fixed before data collection (Nosek et al., 2018) for 15 countries and then extended to 25; the plan was never deposited publicly, so we do not claim pre-registration, and the study is reported as exploratory throughout.

Contributions of this work

1. *An auditing protocol for regulatory information, not a trivia benchmark.* The question an administration faces is not whether a model writes well about policy but whether the value it returns matches the official register. We build five task types from the everyday work of an environmental agency (Table 1), each keyed to an official source, so correctness is adjudicable rather than a matter of expert impression, and the protocol transfers to any regulated domain where the correct answer is published.
2. *The three localisation fixes agencies reach for, tested rather than assumed — and all three fail.* Persona framing, native-language prompting and a regionally tuned model are each a procurement or workflow decision, and each circulates as received wisdom. None works, and the one a manager is most likely to try first does active harm: asking in the country's own language lowers accuracy by 4.8 percentage points, most sharply for the language with the scarcest corpus.
3. *A design that separates who is poorly served from why.* Countries are stratified on independent axes — the UNCTAD developed/developing split, the Joshi linguistic-resource taxonomy, and development — so

geographic, linguistic and economic explanations can be told apart. Between countries, coverage and development are collinear and cannot be separated; they separate once the unit of variation becomes the task within a country, which identifies country coverage as the operative channel. We also flag that the pre-specified corpus proxy is defective in a way worth knowing for anyone who has used it: it assigns the English Wikipedia to every sample country with English as an official language, indexing colonial history rather than corpus size.

4. *Ground truth adjudicated by code, and what that changes.* Rebuilding two task keys from official registers and moving range comparison into code cut the development gradient from $\rho = 0.51$ to $\rho = 0.37$ and more than doubled the native-language penalty, on the same responses. In this literature the ground-truth key, not the sample size, is the binding constraint.

5. *Open resources.* Protocol, prompts, ground-truth registries with their official sources, analysis code, and the deviation log are released under CC-BY-4.0 and MIT, with a one-command reproduction pipeline.

2. Related Work

Prior work runs along four threads: the measurement of geographic factual bias, multilingual and cultural extensions, mitigation attempts, and LLM use in policy and health decision contexts.

2.1. Geographic factual bias and multilingual extensions

Manvi et al. (2024) established that frontier LLMs carry systematic biases against regions of lower socioeconomic conditions, with correlations up to 0.70 between model-generated ratings and a proxy for those conditions — though the outcome space is subjective ratings, which leaves open whether the same bias appears in objective recall of regulatory facts. Moayeri et al. (2024) quantified factual recall against World Bank indicators across 20 LLMs, documenting 1.5× higher absolute relative error for Sub-Saharan Africa than North America; its design is bounded by the macro-economic indicator space, while an environmental manager asks narrower, source-specific questions whose ground truth lives in environmental councils and monitoring agencies rather than one aggregated database. Mirza et al. (2024) showed that newer GPT-4 versions do not monotonically improve and that the privileging of Global North statements persists.

On the multilingual side, Myung et al. (2024) covered 16 countries and 13 languages in everyday cultural knowledge, finding that LLMs perform better in the native languages of high-resource cultures and worse in those of low-resource ones — the finding that motivates H2 and connects to the resource hierarchy of Joshi et al. (2020). Regional benchmarks have proliferated: BRoverbs (Almeida et al., 2025b), TiEBe (Almeida et al., 2025c), ALBA (Vieira et al., 2026) and AMALIA (Simplício et al., 2026) document concrete failures in culturally grounded tasks but do not address applied policy use. Work on Portuguese corpus construction is explicit that the effort has centred on English and that building effective corpora for other languages remains open: Almeida et al. (2025a) assemble a 120B-token Portuguese corpus precisely because language-specific pipelines are needed to reach industrial quality. That asymmetry in how much text exists per language is what H4 tests.

These studies establish that the phenomenon exists and is measurable. None isolates a single applied domain with verifiable official ground truth, and none treats prompt framing as a manipulable factor.

2.2. Mitigation, persona, and role conditioning

Opuszko and Böhm (2026) found that reasoning helps in aggregate but does not eliminate regional disparities, and Kerche et al. (2026) proposed a place-based typology situating geographic bias in a broader sociotechnical landscape. Both are conceptual and aggregate.

Role conditioning is not a folk practice but what the vendors instruct their own users to do: Anthropic documents role setting in the system prompt (Anthropic, 2026), OpenAI's guide opens the recommended message structure with an identity block (OpenAI, 2026), Google's Gemini documentation asks that role definitions — which it calls persona — be placed in the system instruction (Google, 2026), xAI's standard example initialises every conversation with a role-assigning system prompt (xAI, 2026), and the pattern is catalogued in the prompting literature (White et al., 2023). The systematic evidence on whether it helps is sobering: across four LLM families and 2,410 factual questions, expert personas did not improve accuracy over a no-persona control and slightly *reduced* it across 162 roles (Zheng et al., 2024). An independent study with a different design — six models on graduate-level multiple-choice questions — reaches the same conclusion: matching an expert persona to the problem type had no significant effect on performance (Basil et al., 2025). Whether persona conditioning helps or harms factual reliability in high-stakes domains is therefore unsettled and, to our knowledge, untested for geographic equity. We treat persona as a pre-specified experimental factor and test both the Norm-Compliance hypothesis (persona reduces the gap) and the Persona-Vacuum hypothesis (persona amplifies it).

2.3. LLMs in policy, health, and the administration

Le Coz et al. (2025) evaluated LLM policy recommendations for homelessness across four cities — Barcelona, Johannesburg, South Bend and Macau — and read the pattern as "a context-blind rigidity and a potential bias based on the published online discourse on homelessness in the regions of the world with overrepresented data in LLMs", against experts who tailored decisions to locally encoded realities. That is one domain and one task type, which we generalise to five task types spanning recall, synthesis and recommendation. In health, Kozlakidis et al. (2026) argue that hallucinations carry acute risk in regulatory environments with limited technical capacity, where language and cultural mismatch increases error, a risk that meets the mortality burden documented by the Global Burden of Disease estimates (GBD 2019 Risk Factors Collaborators, 2020).

The deployment question is no longer hypothetical. Kim (2026) estimates bureaucratic productivity effects of generative AI quasi-experimentally, and Robinson (2026) documents that choosing which model an agency runs is now an explicit procurement decision between open-weight and vendor-hosted options — a decision both our model-level findings speak to, since the openly available frontier models trail the closed ones and the regionally tuned Portuguese model is the weakest of the fourteen. Two further strands frame why a wrong answer matters administratively: Choroszewicz (2026) shows that generative-AI support tools position frontline practitioners as "moral crumple zones", absorbing responsibility for failures originating upstream, and

Cordella and Gualdi (2024), analysing the first binding measure a regulator imposed on a generative-AI service — Italy's intervention on ChatGPT — argue that technology-neutral frameworks overlook what is specific to these systems and that regulators must engage their technical workings directly. On reception, Aoki et al. (2024) test whether the type of explanation changes perceived accuracy, fairness and trustworthiness among those affected.

What none of this establishes is whether the underlying answer is *correct*, and whether correctness is distributed evenly across the states that rely on it. Perceived trustworthiness and measured accuracy are different quantities, and a system can be well explained, well governed, well received, and wrong. That is the gap this study addresses, in a domain where the correct answer is published in an official register and can be checked.

2.4. Positioning of the present work

Our design differs from prior work in four ways: (i) a single high-stakes applied domain with verifiable official ground truth, rather than numeric trivia or cultural knowledge; (ii) persona as a pre-specified within-subject factor rather than an uncontrolled framing choice; (iii) theory-driven country sampling on independent axes (UNCTAD coloniality, Joshi linguistic resource, development) rather than regional convenience; and (iv) an explicit corpus-representation mechanism test with sensitivity analysis. No prior study combines a regulated public-health domain, a persona manipulation, and a pre-specified mechanism test.

3. Methods

3.1. Research design

We specify a pre-specified, factorial, multi-country benchmark experiment quantifying geographic and persona-modulated bias in 14 large language models on air pollution policy tasks. The pre-specified design crosses 15 countries (stratified on three theoretically grounded axes), 14 LLMs spanning five access tiers, one domain, 5 task types and 2 persona conditions, with 2 stochastic replications per cell; a post-hoc extension (Section 3.3) enlarges the sample to 25 countries. The stimulus set crosses each country's five tasks and two personas in English, plus a within-country native-language rendering for the nine countries with a target native language, scored on the $[0, 1]$ composite. The protocol fixes, in advance of collection, the hypotheses, primary models, persona protocol, exclusion criteria and multiple-comparison corrections. A Brazil-only collection executed under an earlier three-domain design is reclassified as an extended pilot and reported separately; it informed prompt calibration but contributes no confirmatory observations.

3.2. Country selection

Countries were selected by theoretical sampling (Patton, 2015) triangulated across three independent classifications: the Global North/South distinction, operationalised through UNCTAD's developed/developing economies classification (UNCTAD, 2024), which carries the coloniality-of-knowledge framework motivating the study (Mohamed et al., 2020); the Joshi et al. six-class taxonomy of linguistic-resource representation (Joshi et al., 2020); and World Bank income groups (World Bank Group, 2025).

The pre-specified sample of 15 — Brazil, Mexico, Argentina, Peru; Nigeria, South Africa, Kenya, Egypt; India, Indonesia, Bangladesh, the Philippines; and the United States, Germany, Japan as Global North references — covers four world regions and four income groups. The linguistic-resource axis turned out far less differentiated than intended: the dominant official languages fall into only three Joshi classes, twelve of the fifteen sharing Class 5, and extending to 25 does not help since eighteen are Class 5. This is a property of the world rather than of our draw, because the official languages of most states are themselves high-resource, but it means the linguistic axis has too little variance to be separated from the others, and we treat the Joshi results accordingly (Section 4.6). Countries with compromised statistical governance or without a publicly retrievable national standard were excluded; Oceania and Joshi Class 0 languages are not represented, which we note as a scope limitation.

3.3. Sample expansion after the pre-specified analysis

After the pre-specified 15-country analysis, and *reported here as a post-hoc extension rather than part of the locked protocol*, the sample was enlarged to 25 countries under identical procedures, to raise power for the gradient and mechanism tests and to multiply the within-country language pairs available for H2. The manuscript reports the 25-country sample throughout; the primary family recomputed on the pre-specified 15 alone is tabulated in Supplementary Material, so a reader can see what the locked protocol would have concluded. Because the extension followed observation of the initial results, no pre-specified conclusion is revised on the extension alone, and the expansion is treated as a power-and-robustness check rather than a new confirmatory test. Angola, like Nigeria and Argentina, has *no* national ambient $PM_{2.5}$ standard, adding no-standard cases in which a faithful model must decline to state a non-existent threshold. The ten additions, their roles, and the two extra question variants generated per (country, task) for the H2 pairs are detailed in Supplementary Material.

3.4. Country covariates

Human Development Index (UNDP 2023–24) and GDP per capita index developmental position and enter the mechanism analysis as adjustment covariates. The corpus-representation exposures divide along a distinction the mechanism analysis makes explicit, because the two channels apply to different effects. *Language-corpus* measures quantify how much pretraining text exists *in* a language and are the candidate mechanism for the native-language penalty: mC4 token count (Xue et al., 2021, Table 6), OSCAR-2301 corpus size (corpus introduced by Abadji et al., 2022; per-language figures read from the OSCAR-2301 dataset card, whose totals differ from the 22.01 release described in that paper), and the Joshi class (Joshi et al., 2020). *Country-corpus* measures quantify how much the encyclopedic corpus is *about* a country, independent of language, and are the candidate mechanism for the geographic gap: the English Wikipedia article length, the number of Wikipedia language editions with an article on the country, and the number of Wikidata statements on the country entity.

The two families are not interchangeable. Because the geographic gap is measured *in English*, with language held constant across countries, a residual North/South gap cannot be a language-corpus effect. One point deserves stating plainly, because the roles are easy to conflate: Wikipedia and Wikidata are never ground truth here, they are the *explanatory* variable. What an answer is scored against is always an official register (Section 3.9); had we scored against an encyclopedia, the study would measure agreement with an encyclopedia, which is not the question a public administration faces.

3.5. Model selection

Fourteen LLMs were benchmarked across five access tiers, spanning roughly two orders of magnitude in parameter count and four training-data geographies (United States, China, Europe, Brazil): open-weight frontier (5), open-weight mid (3) and open-weight small or regional (2) run locally under pinned tags, and closed-accessible (2) and closed-frontier (1) via official APIs. The small tier includes a Brazilian-Portuguese instruction-tuned model (Cabra-Mistral 7B v3), tested against a scale-matched globally trained open model for H3. Each model was queried with an identical pinned tag throughout the collection window. The full roster is in Supplementary Table S1.

We compare *models as served* through a fixed access stack, not idealised weights. Provider infrastructure, serving configuration and API layer are part of what each label denotes, and a residual gap could in principle reflect the stack rather than the weights; we hold this constant by pinning version tags and the maximum-determinism configuration available per provider throughout a fixed window.

3.6. Domain and task taxonomy

The single domain is air pollution policy, chosen because it anchors to officially published, verifiable ground truth and maps to concrete decision activities of a public environmental manager. The five tasks were not written as a generic knowledge quiz: each corresponds to a question a manager has to answer in the course of ordinary work, and each was drafted so that a wrong answer would carry an administrative consequence rather than merely lose a point. Table 1 gives them with the prompt exactly as issued to the models, using Brazil as the worked instance, alongside the decision each stands in for.

The tasks are ordered by how much of the answer is fixed by an official source. T1, T2 and T4 admit verifiable ground truth and carry the primary accuracy signal; T3 and T5 are rubric-scored open-generation tasks, where competent answers legitimately differ. That ordering is what makes the task dimension informative rather than decorative: it lets us ask whether models fail uniformly or fail specifically where a published national value has to be recalled, which is the question the results turn on. The country-by-task mapping of official source classes and verifiability status is in the ground-truth registry (Section 3.9), and the full 25-country prompt set is released with the data.

3.7. Persona manipulation

Persona is a within-prompt factor with two levels, and the manipulation is not arbitrary: assigning a role to the model is the practice the vendors themselves document and recommend (Section 2). The neutral condition presents the query as a plain information request; the public_manager_env condition prepends a fixed role frame identifying the requester as a municipal or regional environmental management official, holding the substantive question identical. For Brazil's T1 the frame is the single sentence *"I am the municipal environmental management official of São Paulo, Brazil, and I* *need to brief decision-makers."* A single template is applied uniformly across countries and tasks, translated where the prompt language is non-English, so any between-condition difference is attributable to the frame rather than to question wording. Persona order within each prompt-model cell is randomised, and a manipulation check verifies that the frame is acknowledged in the response. Exact wording, translation procedure and manipulation-check coding are in Supplementary Material.

One implementation detail matters for how the result should be read. Every request in this study is sent as a single user turn with no system prompt, so the role frame is prefixed to the user message rather than installed as a system instruction. Three of the four vendor guides place the role there; Google's is explicit that it belongs *either* in the system instruction *or* "at the very beginning of the user prompt" (Google, 2026), which is exactly the channel we test. This is deliberate: the system-prompt route requires API access and a developer to configure it, while the public servants whose decisions motivate this study reach these models through the consumer chat interface, where the only available channel is the message they type. We therefore test role framing as the sector actually performs it, and Section 5.7 states what remains untested. Persona is the basis of H6.

3.8. Prompt construction and validation

Prompts were authored for the domain by the research team with domain supervision and validated against the ground-truth registry before entering the confirmatory stimulus set, with every factual claim in a registry entry required to resolve to an official source URL or a peer-reviewed DOI before the corresponding prompt was admitted. The analysis plan called for a 50-

prompt pilot scored by two independent human raters ($\alpha \geq 0.70$) before scaling the design. *That pilot was not run*, and no rater-level α exists for this study; rubric reliability is evidenced instead on the confirmatory data itself, by a three-vendor panel over every item that required a judge (Section 4.11), which returns $\alpha = 0.527$. We report this as a departure from the plan rather than as a threshold met.

All prompts were authored in English and, for a stratified subset of countries, additionally in a relevant native language (Supplementary Section S3). Native-language renderings were produced by an LLM translator (Claude Sonnet 4.6) prompted to preserve meaning, persona frame, proper nouns and acronyms, with the factual ground truth held in English since the target value is language-invariant. We did not employ human native-speaker translators, which remains a limitation (Section 5.7); we did audit the renderings after the fact by back-translating all 90 native prompts with a different model and having two further models compare each against the English original (Section 4.4).

3.9. Ground truth and the ground-truth registry

Ground truth for the verifiable tasks was drawn from official sources of record: national air quality standards and the agencies issuing them (T1), environmental-agency monitoring reports (T2), and federal policy programs with their executing agencies and legal instruments (T4). For T3 the reference is a curated set of peer-reviewed epidemiological studies with resolvable DOIs rather than a single number.

Because the validity of the geographic gradient depends on ground truth being equally available and equally verifiable across countries, we built a per-country registry recording, for each (country, task) pair, the class of equivalent official source, the resolvable URL, the gold value or rubric key, the reference date, the human validator and a validation flag. Entries are versioned by access date, source URL and a SHA-256 hash of the retrieved document. Entries whose official source is unavailable or non-equivalent for a country are flagged and excluded from the primary comparison for that cell, so missing ground truth is never scored as a model error. The registry design and cross-country equivalence criteria are in Supplementary Material.

3.10. Data collection

Calls were executed at temperature 0.3 with deterministic seeds where available, with 2 replications per prompt-model-persona combination, inside a fixed window to minimise model-update confounds; the reported accuracies are therefore point-in-time estimates. Every response was stored with complete request metadata — pinned tag, timestamp, token counts, finish reason, persona condition — under SHA-256 checksums. Quality gates excluded truncated responses and responses in a language divergent from the request, both retained and analysed separately.

Per-cell coverage is not uniform: free and rate-limited endpoints returned fewer admissible responses than metered APIs, so the scored N ranges from roughly 300 to 780 per model. We report each model's realised N alongside its score (Table 2) rather than silently dropping under-covered cells, and the full coverage matrix, with quota and exclusion reasons for every shortfall, is in Supplementary Material.

3.11. Measures

Scoring assigns each task the most reliable instrument it admits.

Where the answer is a number in an official register, code decides. For T2 and T3 the returned value is extracted and compared with the authorised range by a deterministic scorer, with plausibility guards that reject implausible extractions — a four-digit year read as a death count, a national population read as a mortality figure. Comparing a value with a range is exact, and a language model asked to do the same arithmetic can only reproduce it with noise. Code resolves 62.5% of T2 and 83.6% of T3.

Where judgement is required, a three-vendor panel decides by its mean. The residual of T2 and T3, with all of T4, is scored by Gemini 2.5 Pro, Claude Sonnet 4.6 and DeepSeek-V3, chosen for vendor diversity so that intra-vendor correlation does not inflate apparent agreement. T1 and T5 retain the original single-judge scores. Reliability is reported as Krippendorff's α, the single-judge ICC$(2,1)$ and, as the operative quantity, the ICC$(2,3)$ of the panel mean, computed on every item the panel scored rather than on a calibration subset. One panel member (DeepSeek-V3) is also an evaluated model, and we test for self-favouring directly. No human gold-standard layer was collected; ground truth is anchored instead to official primary sources (Section 3.9).

The unit of analysis is the scored cell, not the log line. Of the 9,251 scores the judge produced, 951 cells (11.5%) carry more than one score, because the judge was re-run on parts of the set on different dates. Those are repeated measurements of the same item, not independent observations, so we collapse them by their mean, leaving 8,300 cells (6,629 English-prompt). Treating reruns as independent would be pseudo-replication: it inflates n and gives extra weight to whichever cells happened to be reprocessed. The choice is made on methodological grounds and does not depend on the result; Supplementary Material reports every headline effect under both conventions, and the study's central finding is indifferent to it (-4.755 against -4.753 pp).

The primary composite combines five pre-specified subcomponents (Supplementary Section S6), each normalised to $[0, 1]$: factual accuracy against the registry entry (weight 0.30), contextual completeness (0.25), citation quality (0.15), calibration (0.15) and hallucination absence (0.15). Every headline effect is recomputed under equal weights and a PCA-derived weighting, and a conclusion counts as robust only if its direction is stable across all three. The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test.

3.12. Analytical strategy

The three primary tests are pre-specified as a single family (F1) and corrected together with Bonferroni-Holm. Secondary tests (F2) are reported unadjusted within family, and declared exploratory tests (F3) use Benjamini-Hochberg FDR at $q = 0.10$ (Benjamini and Hochberg, 1995), whose guarantee assumes independent test statistics.

H1 (geographic gradient, primary) regresses country-level mean composite accuracy on the Joshi/HDI gradient via Spearman ρ, pre-specified one-sided at $\rho \geq 0.55$, with a Mann-Kendall trend test as a mandatory non-monotonicity complement. *H4 (corpus mechanism, primary)* relates corpus representation to accuracy by partial Spearman ρ adjusting for HDI and log GDP per capita, with Wikipedia counts as the pre-specified proxy and

Wikidata coverage as exploratory complements. *H6 (persona, primary)* is a country-level difference-in-differences, $\Delta_{\mathrm{GS}} - \Delta_{\mathrm{GN}}$, supported if the persona condition reduces the Global South gap by at least 0.05 on the composite, with inference by a 5,000-permutation country-level test. *H3 (secondary)* is a one-sided Mann-Whitney contrast (Cliff's δ). *H2 and H5 (exploratory)* are reported descriptively under F3 correction.

Three cross-cutting analyses corroborate the tier gap: a Gaussian linear mixed model (`statsmodels` MixedLM, REML) with crossed random intercepts for country and model, adjusting for centred HDI; a Bayesian hierarchical re-estimation (`pymc` (Abril-Pla et al., 2023), weakly informative priors); and E-values for robustness to unmeasured confounding, via the approximation RR $\approx \exp(0.91\, d)$ that VanderWeele and Ding (2017) prescribe for standardised effects and whose assumptions it inherits. The corpus mechanism is probed with an exploratory mediation model (`semopy` (Igolkina and Meshcheryakov, 2020)), and the persona manipulation with a role-frame-acknowledgement check. Pre-specified robustness analyses appear in Section 4.12.

3.13. Study protocol and ethics

The analytic dataset, after quality gates but before model fitting, is frozen and hashed, so the confirmatory analysis can be audited against the locked specification rather than taken on trust. Release terms are stated at the end of the paper.

No personal data from end-users of LLMs were collected; the study analyses model outputs to publicly relevant policy questions. Provider terms of service were reviewed and respected, and responses are redistributed solely for research under fair-use and reproducibility provisions, excluded from any commercial application.

4. Results

Sample note. Results below are reported on the *25-country* sample. The pre-specified benchmark covered 15 countries and was extended post hoc by ten (six Global North references — the United Kingdom, Canada, Australia, South Korea, France, Italy — and four native-pair countries — Colombia, Chile, Portugal, Angola). Fourteen LLMs are scored across five tasks, two personas, and four languages (English plus a country-native language — Spanish, Portuguese, or Hindi — for the nine applicable countries, giving the within-country English/native pairs that test H2). The primary family recomputed on the pre-specified 15 countries alone is tabulated in Supplementary Material, and no pre-specified conclusion is revised on the extension alone; where the two samples differ on a hypothesis the paper turns on, we say so in the text. Scoring is adjudicated by code wherever the answer is a number checkable against an official register, and by the mean of a three-vendor judge panel elsewhere (Section 4.11); the T1 ground truth is anchored to official primary sources for all 25 countries (Section 4.9).

4.1. Sample and scoring

The analysed corpus comprises 8,300 scored cells on the $[0, 1]$ composite, of which 6,629 are English-prompt responses across the 25 countries; the remainder are the within-country native-language pairs used for H2. Scoring uses the most reliable instrument each task admits: code adjudicates against the official register wherever the answer is a published number, resolving

62.5% of T2 and 83.6% of T3, and the residual, with all of T4, is scored by the mean of a three-vendor panel. T1 and T5 retain their original scores. In total 5,373 cells carry a corrected score.

The panel mean replaces the single-judge design of the original protocol, in which the judge (GPT-5-mini) was itself among the evaluated models. Judges differ sharply in severity (mean composite 0.617 for Gemini, 0.451 for Claude, 0.379 for DeepSeek), so a single judge fixes an arbitrary level, and single-judge reliability is only $\mathrm{ICC}(2,1) = 0.56$ against $\mathrm{ICC}(2,3) = 0.79$ for the panel mean. One panel member (DeepSeek-V3) is also an evaluated model, so we checked for self-favouring: it scores its own outputs *more* harshly relative to the other two judges (-0.177) than it scores any other model's (-0.123 to -0.160).

4.2. Accuracy by model

Table 2 orders the 14 models by mean composite: frontier systems lead and small open-weight models trail. On H3, the Brazilian-Portuguese regional model Cabra-Mistral 7B is the *weakest* of all (0.347 versus 0.548 for the rest; Mann-Whitney $p \approx 0$, Cliff's $\delta = -0.47$). Cliff's δ has a direct reading — the probability that a randomly drawn answer from one group beats one from the other, rescaled to $[-1, +1]$ — so -0.47 means the regional model loses roughly three such comparisons for every one it wins. Scale and frontier training dominate regional instruction tuning at the 7B level.

4.3. Accuracy by task: the recall floor

Difficulty is sharply task-dependent (Table 3). The hardest task by a wide margin is T1 (technical-standard recall, 0.369) — the task that requires returning the national annual $PM_{2.5}$ standard in force. The open recommendation task (T5) scores highest (0.769) because its rubric rewards plausible, well-structured policy advice that does not hinge on a single fact. Pooling the two recall tasks (T1, T2; mean 0.421) against synthesis and recommendation (T3–T5, 0.612), the difference is large (Mann-Whitney $p \approx 0$, Cliff's $\delta = -0.47$, again roughly three losses for every win). LLMs are weakest exactly where applied environmental management needs them most: recalling the precise regulatory threshold in force.

The correction of the ground truth is itself informative here, and provides an internal check on it. T1 always had official ground truth, and its score barely moves (0.367 to 0.369). T2 was previously scored against a placeholder, and it rises from 0.368 to 0.473 once each returned concentration is compared with the authorised range by code. The correction therefore moved exactly the task whose ground truth was defective and left untouched the task whose ground truth was sound, which is the behaviour expected if the correction is right. It also revises the earlier reading: the floor is real but shallower than reported, and it is concentrated in normative recall (T1) rather than shared equally with local-datum recall (T2).

4.4. H2 — asking in the local language lowers accuracy

For the nine countries with a target native language, each prompt is rendered in both English and the native language with content held fixed, so the contrast is within-country and within-model. Replicates are averaged within each (model, prompt) cell before pairing, since pairing replicate-to-replicate would inflate n and make the test anticonservative. Across the 839 matched cells, native-language prompting *lowers* mean composite accuracy by 4.8 percentage points (Wilcoxon signed-rank $p = 3 \times 10^{-15}$). The penalty tracks the resource level of the language, and all three are significant (Table 4): Hindi -11.1 pp, Spanish -4.4 pp, Portuguese -3.9 pp. India, which scores highest of all 25 countries in English, drops the furthest in Hindi.

Because H2 carries the study's central claim, we tried to make it disappear rather than to confirm it, and it held under every attack: it is not driven by any one country (leave-one-out -4.2 to -5.2 pp, all significant), by any one model (-4.2 to -5.1 pp), or by outlying cells (5% trimmed -4.3 pp, $p = 5 \times 10^{-19}$); it survives every weighting of the composite; and it appears in every task and under both persona conditions.

Two attacks answer the objections a sceptical reader raises first. The first is that language-model judges might penalise non-English prose for style rather than content, so we test a binary outcome no judge touches: does the response contain a value the extractor can compare with the register? English prompts yield a verifiable value 66.4% of the time and native-language prompts 48.7%, a paired difference of -17.7 pp, with the native version worse in 122 of the 182 discordant pairs (sign test $p = 5 \times 10^{-6}$) and the same ordering by language. Asking in the local language does not merely lower a judged score; it makes the model substantially less likely to return a checkable number at all.

The second is that the native prompts, produced by an LLM translator, might simply ask something different. We tested this on all 90 unique native prompts — a census, not a sample — back-translating each with a *different* model and having two further models audit the result against the English original. On the two items that could confound the effect, fidelity is complete:

no prompt changed the country and none changed the requested quantity (90/90 on both). Restricting H2 to the prompts verified as faithful leaves it essentially unchanged (-4.5 pp, $p = 9 \times 10^{-8}$) against -5.2 pp among the divergent ones, a difference not distinguishable from zero (permutation $p = 0.42$). Translation fidelity does not predict the size of the penalty, which is what one expects if translation is not causing it. The effect is weakest in the code-adjudicated subset (-2.2 pp, not significant) for a structural reason: code adjudicates only where a number exists, so that subset conditions on the model having produced a value in *both* languages, selecting away the severest failure mode.

The applied implication is the opposite of the intuition. A manager who suspects an English-language model serves their country poorly, and responds by asking in the national language, makes the answer *worse* — and worse by the largest margin precisely where the language is least represented in training corpora.

4.5. H6 — persona does not narrow the geographic gap

The persona factor was fully crossed, so H6 is tested as a difference-in-differences. The public-environmental-manager frame leaves the North/South gap essentially unchanged: $+5.0$ pp under the neutral frame versus $+5.9$ pp under the manager frame, a DiD of $+0.8$ pp, far below the pre-specified 5 pp threshold; a country-level permutation test (5,000 permutations of the tier label) gives $p = 0.29$. H6 is not supported: presenting the query as coming from a public environmental manager neither narrows nor amplifies the Global South disadvantage, consistent with evidence that expert personas do not reliably improve factual accuracy (Zheng et al., 2024; Basil et al., 2025). A manipulation check finds that 50.2% of persona-condition responses explicitly acknowledge the role frame, so the null is not an artefact of the frame being universally ignored.

4.6. H1 — a tier gap that holds, a gradient that does not

H1 was pre-specified in two parts, and they separate cleanly.

The ten Global North countries average 0.568 against 0.514 for the fifteen Global South countries, a gap of +5.4 percentage points whose 95% bootstrap confidence interval over countries is $[+2.1, +8.7]$ pp and *excludes zero*. It is stable to dropping any single country (leave-one-out $[+4.9, +6.4]$ pp) and to the composite weighting (+4.2 pp under equal weights, +8.8 pp on the factual subcomponent, where it widens).

Where the gap lives, and why the composite understates it. The composite averages five tasks and the gap is not spread evenly across them. Re-estimating the same contrast as a binomial mixed model on each task separately — outcome coded as returning the register value, with crossed random intercepts for country and model — shows the disadvantage scaling with how much the task depends on a fact specific to that country (Table 5). On the binding national standard (T1) the Global South odds of a correct answer are 0.22 of the Global North's (52.2% versus 29.4% raw), and on policy instruments (T4) 0.33; on health-evidence synthesis (T3), where the answer is an international estimate rather than a national one, the ratio rises to 0.69; and on open recommendation (T5), the one task with no register value to get right, the gap disappears entirely. The gap is not weak — it is *concentrated*, and the composite dilutes it by averaging tasks where it is severe with a task where it does not exist. On the question a public manager most often brings to these systems, what is the standard in force here, a Global South country is served at roughly a quarter of the odds of a Global North one.

At $n = 25$, Spearman $\rho(\text{accuracy}, \text{HDI}) = +0.41$ clears conventional significance ($p = 0.043$; Mann-Kendall $p = 0.042$) but stays well below the pre-specified $\rho \geq 0.55$ threshold, which is the criterion H1 was registered against. At the pre-specified $n = 15$ it is essentially absent ($\rho = +0.10$), and no leave-one-country-out re-estimate reaches 0.55 either (range $[+0.35, +0.54]$). We therefore report a tier difference we can demonstrate and a development gradient we cannot: the disadvantage is established as a difference between groups, while its shape across the development range remains exploratory. Under Bonferroni-Holm across the primary family {H1-gradient, H4, H6}, no test rejects its null.

The country ordering retains large axis-orthogonal heterogeneity (Supplementary Material): India (0.652, Global South) tops all countries, South Africa and Indonesia outrank most of the Global North, and Canada and France sit low. The disadvantage is a tier difference layered over substantial country-specific variation, and that heterogeneity is itself part of why the gradient does not resolve.

4.7. H4 — corpus representation: country coverage, not language size

The pre-specified mechanism was training-corpus representation, proxied by Wikipedia article and edition count. That proxy is null: Spearman $\rho = +0.14$ ($p = 0.50$) uncontrolled, and still null partialling out HDI and log GDP per capita. A post-hoc country-coverage proxy reaches $\rho = +0.40$ ($p = 0.048$) and attenuates after adjusting for HDI (partial $\rho = +0.19$,

$p = 0.36$). At country level neither channel is established, because with 25 countries coverage and development are collinear.

Changing the unit of variation resolves what the country level cannot

The obstacle is confounding rather than absence of signal, and the way out does not require more countries. Each country answers five tasks that differ in how much the answer depends on a fact about *that* country: T1, T2 and T4 do; T3 (an international WHO estimate) and T5 (a generic recommendation) do not. Countries score 0.454 on the first block and 0.658 on the second, a specificity deficit of 0.204. If what limits the model is how much the corpus says about a given country, that deficit should be smaller where coverage is greater — and a country's HDI cannot explain a deficit measured *within* that same country. Estimating the interaction between coverage and task dependence at the response level, with a fixed effect for country that absorbs development, income, official language and everything else distinguishing countries, gives $\beta = +0.028$ (SE 0.005, $p = 7 \times 10^{-9}$, $n = 6{,}629$).

Four checks separate mechanism from artefact (Table 6). The *purest* contrast — T1, wholly national, against T5, wholly generic — yields the *largest* coefficient, the dose-response a real effect produces and a spurious association does not. Dropping T5 or T4 leaves it intact, and a placebo inverting the labels returns the exact mirror image. Entering both candidate channels together, country coverage remains strong ($\beta = +0.027$, $p = 2 \times 10^{-8}$) while the language-corpus proxy does not survive ($\beta = +0.008$, $p = 0.089$). That prediction was asymmetric and therefore falsifiable: because every response here was elicited *in English*, the size of a country's local-language corpus has no obvious route to how much the model knows about that country, so had the language channel survived and coverage failed, our reading would have been wrong.

The resulting claim is narrower than "coverage causes the gap". The design removes country-level confounding by construction, not confounding at the level of task-by-country. What we can say is that the channel makes a differential prediction, that it holds, that it strengthens as the contrast sharpens, and that it survives adjustment for the rival channel.

The corpus account also shows itself in H2, where the unit is the language: the per-language penalty falls monotonically with mC4 token volume (Xue et al., 2021) and OSCAR size (Abadji et al., 2022), with Hindi, the smallest corpus, carrying the largest penalty — though with three languages this remains descriptive.

4.8. H5 — open versus closed-accessible

Closed-accessible models average +12.6 pp above open-weight ones, against the optimistic H5 framing. The open arm includes larger models, so the gap is not attributable to scale alone, although DeepSeek-V3 ranks first overall. H5 is interpreted descriptively.

4.9. Ground-truth provenance

Every key comes from an official register, anchored by documentary verification and reproducible from the cited URLs: T1 from the national regulation, T2 from WHO Ambient Air Quality Database v6.1, T3 from WHO GHO indicator AIR_41, T4 from the UNEP Global Assessment of Air Pollution Legislation. Three countries have *no* national $PM_{2.5}$ standard, confirmed from the official text — Nigeria regulates PM_{10} only, Argentina has no binding federal value, Angola no national legislation — which is where a faithful model must decline to state a non-existent threshold. The full registry is in Supplementary Material. Wikipedia and Wikidata appear only as the explanatory variable of the mechanism test; no answer is scored against them.

4.10. Qualitative failure modes

Three failure modes recur behind the composite, with verbatim examples in Supplementary Material. Models *fabricate standards that do not exist* for the three countries with no national norm, inventing a plausible threshold rather than abstaining and disagreeing among replicates about the invented value; they *miss the binding value*, answering 15 μg/m^3 for Brazil where the standard is 17 (CONAMA 506/2024); and they *degrade in the native language*, as when GPT-OSS 120B returned India's correct 40 μg/m^3 in English and an *empty* answer in Hindi.

4.11. Inter-judge reliability

Reliability is reported on the sample that produces the effects, not on a calibration subset. All 3,190 items requiring a judge were scored by all three panel members, giving Krippendorff's $\alpha = 0.527$, single-judge $\mathrm{ICC}(2,1) = 0.558$, and — the operative quantity, since the analysis uses the panel mean — $\mathrm{ICC}(2,3) = 0.791$.

The panel also isolates a design rule that generalises beyond this study. Where the judging prompt asked the model to compare a returned concentration against a tolerance band expressed in words — that is, to perform the arithmetic itself — inter-judge concordance on those items was $r = -0.04$.

Pre-computing the admissible range and asking only whether the value falls inside it raised concordance on the same items to $r = +1.00$, and the 274 scores collected under the ambiguous wording were discarded and recollected. Apparent judge disagreement can be an artefact of what the judge is asked to compute. Arithmetic belongs to code.

4.12. Robustness

The two supported effects were tested against seven attacks and survive all of them; the development gradient fails all of them in the same direction, which is what makes its exploratory status a finding rather than an omission. Full outputs are in Supplementary Material.

No single country, and no weighting, carries the result. Leave-one-country-out re-estimation holds the tier gap in $[+4.9, +6.4]$ pp and the native-language penalty at -4.7 to -5.0 pp. Under equal weights the gap is $+4.6$ pp and the language penalty -4.7 pp; restricting to factual accuracy against the register value widens them to $+9.4$ and -5.4 pp and deepens the recall floor, so both effects grow where the measurement is most objective.

The tier gap survives a distribution-free test and two model families. A permutation test shuffles the North/South labels among the 25 countries ten thousand times and counts how often chance alone produces a gap this large; it assumes nothing about the distribution and treats the country, not the answer, as the independent unit. Chance clears the observed gap 2.0% of the time ($p = 0.020$). A Bayesian hierarchical model with crossed random intercepts agrees ($\beta = -0.052$, 94% HDI $[-0.092, -0.016]$, $P(\beta < 0) = 0.994$), while a frequentist mixed model returns the same magnitude and sign but leaves it marginal ($\beta = -0.066$, $p = 0.069$), because it charges the large between-country heterogeneity to the error term. Two of three tests clear zero and all three agree on magnitude, so we report the gap as supported and specification-sensitive, with an E-value of 1.70 at the point estimate and 1.31 at the confidence limit nearest the null (VanderWeele and Ding, 2017) — a country-level confounder of modest strength would suffice to explain it away.

The gap tracks world-system position, not national wealth. Re-estimated under three development-based partitions of the same 25 countries, the gap is significant under UNCTAD and non-significant under every split drawn on HDI (Supplementary Material). This coheres with the rest of the paper: the HDI gradient is itself null, so a split on HDI should not be expected to produce a gap, and that UNCTAD separates the data where HDI does not points to position in the world system rather than income as the operative variable (Quijano, 2000; Mohamed et al., 2020). The reading is post hoc, so we claim the gap for the North/South distinction as conventionally drawn, not as a general statement about development.

What the null gradient does and does not exclude. Simulating 25-country samples at known correlations, the design has 77% power at $\rho = 0.55$ — the threshold fixed in advance — and a minimum detectable effect of $\rho = 0.56$. A gradient of the magnitude we committed to calling substantial is excluded; a weak one is not (power 48% at $\rho = 0.40$). The ten countries added post hoc fall *within* the HDI range the original fifteen already spanned, so the extension raises the density of observations rather than the range of the predictor.

4.13. Summary

Table 7 maps each hypothesis to its finding, the candidate mechanism, and an explicit evidence status, so no claim is read as stronger than the data support. Three effects are supported and survive every robustness attack in Section 4.12; two pre-specified expectations are not met; and the corpus mechanism is identified within countries, where the design has resolution, and not between them.

5. Discussion

We set out to measure whether LLMs answer air pollution policy questions as reliably for the Global South as for the Global North, whether prompt framing changes the answer, and what mechanism drives any gap. Across 25 countries, 14 models, four languages, five tasks and two personas, scored against official ground truth by code where the answer is a register value and by a three-vendor panel elsewhere, the harms we can demonstrate are practical rather than aetiological: the three remedies an agency would reach for all fail, the most intuitive of them makes the answer materially worse, and models are least reliable on the recall of the binding value in force. A North/South tier difference persists, and we identify the channel behind it only where the design has the resolution to do so.

5.1. Three intuitive remedies, all failing — and one that harms

The local language does not help and materially hurts. Asking in the country's own language lowers accuracy by 4.8 pp, significantly in all three languages, and the penalty scales with the scarcity of the language's corpus, with India — the highest-scoring country of all 25 in English — falling furthest in Hindi. This is the mechanism the resource hierarchy predicts (Joshi et al., 2020; Xue et al., 2021), and the measurement has the resolution to see it because the unit is the language rather than the country. The effect depends on no single country, model, task, persona condition or scoring instrument, nor on translation quality, and it holds on an outcome involving no judge at all: the model becomes markedly less likely to produce a checkable number in the first place.

The local-manager persona does not narrow the gap (DiD +0.8 pp, permutation $p = 0.29$), consistent with evidence that expert personas do not reliably improve factual accuracy (Zheng et al., 2024; Basil et al., 2025). This null carries more weight than a null about a folk habit would, because role framing is the first thing every major vendor tells its users to do — Anthropic, OpenAI, Google and xAI converge on the same advice (Section 2), so an administration following its supplier's documentation writes the sentence we tested. What that documentation claims for the sentence is control over tone, format and framing, and nothing in it marks the boundary: nothing tells a reader that the technique which reliably shapes register does not also shape factual reliability. Our estimate separates the two. A manipulation check confirms half the responses explicitly take up the role, so the frame is received and changes the voice; it does not change what the model knows. We tested it typed into the message rather than installed as a system instruction, the channel a public servant in a chat interface actually has (Section 5.7).

The regional model is the *worst* performer of all fourteen ($\delta = -0.47$), so reaching for a locally tuned small model trades away the scale and frontier training that carry domain accuracy.

Practitioners deploy all three as fixes; none works, and the one a manager is most likely to try first does active damage. For Global South environmental agencies, no cheap prompt- or model-level workaround substitutes for verifying the binding value against the official source.

5.2. The recall floor sits on the values that bind

Accuracy collapses on the tasks that require returning a specific value in force: 0.421 for normative and local-datum recall against 0.612 for synthesis and recommendation ($\delta = -0.45$), deepest on the national standard itself (0.369). The risk falls unevenly: a manager drafting prose about air quality policy is comparatively well served, a manager retrieving the regulatory cutoff a decision turns on is in the worst cell of the design, and worse still if they ask in a lower-resource language, and an internal check supports that the floor is genuine rather than an artefact of scoring: correcting the ground truth raised T2 — the task scored against a placeholder — from 0.368 to 0.473, while leaving T1, which always had official ground truth, essentially unchanged (0.367 to 0.369).

5.3. A tier difference we can show, and a gradient we cannot

H1 was pre-specified in two parts, and the corrected scoring separates them: the tier gap holds and the monotonic gradient does not.

Read on the composite the gap looks modest (+5.4 pp, bootstrap CI $[+2.1, +8.7]$). Estimated task by task on the binary outcome of returning the register value, it is not: the gap scales with how much the answer depends on a fact specific to that country, from an odds ratio of 0.22 on the binding national standard to no gap at all on open recommendation, the one task with no register value to get right. The disadvantage is concentrated, and averaging five tasks into one composite dilutes it with tasks where it does not exist. On the question a manager actually brings to these systems — what is the standard in force here — a Global South country is served at roughly a quarter of the odds of a Global North one. That is the faithful statement of the geographic finding.

The gradient fails on every reading, and the null is informative rather than empty because the design has 77% power at the pre-specified $\rho = 0.55$: a gradient of the magnitude we committed to is excluded, while a weak one is not. The country ordering shows why the two results differ. India tops all 25 countries despite being Global South; South Africa and Indonesia outrank most of the Global North; Canada and France sit low. The disadvantage is a tier difference layered over substantial country-specific idiosyncrasy, and that idiosyncrasy is what a rank correlation over 25 countries lacks the resolution to see through.

5.4. The mechanism, where the design has resolution

The pre-specified mechanism test fails, and the reason is a defect in the instrument rather than an absence of signal. The proxy assigns the size of the Wikipedia edition in a country's dominant official language, so nine of our 25 countries receive the value of the *English* Wikipedia. The variable does not measure how much text exists in a country's language; it measures whether the country has English as an official language, which is a fact about colonial history. We report this because it bears on any study that has used or will use the same proxy.

Resolution comes from changing the unit of variation, and the decisive check restricts to those same nine anglophone countries. There the linguistic channel is not merely controlled but identical by construction, and country coverage still predicts the specificity deficit at the full-sample magnitude ($\beta = +0.026$, $p = 7 \times 10^{-4}$). What that coefficient measures cannot be language. Country representation is therefore the operative channel for a gap measured in English, identified in a contrast that removes country-level confounding by construction, but not causally — the design eliminates confounding at the level of the country, not of task-by-country.

5.5. The methodological lesson: the key matters more than the sample

The same 8,300 scored cells support different substantive conclusions depending on the instrument used to score them, and the size of that difference is the lesson. Scoring two of the five tasks against placeholder ground truth, and delegating to a language model the arithmetic of deciding whether a returned number fell inside an authorised range, yields a development gradient of $\rho = 0.51$ ($p = 0.004$) and a native-language penalty of -2.1 pp. Rebuilding those keys from official registers and moving the comparison into code cuts the gradient to $\rho = 0.41$, below the registered threshold, more than doubles the language penalty to -4.8 pp, and converts Spanish from a null into a significant harm. The dataset did not change; only the measuring instrument did. The field's instinct when a geographic-bias effect proves fragile is to add countries, but the ground truth deserves the scrutiny first: a defective key does not merely add noise, it adds *structured* noise, because the defects concentrate wherever official data are hardest to obtain — which is exactly where the hypothesis lives.

5.6. *Why the domain matters*

Fine particulate matter is among the leading environmental risk factors for premature mortality, and the burden falls disproportionately on the Global South: the World Health Organization attributes 88,548 deaths in Brazil in 2021 to ambient air pollution (uncertainty interval 69,477 to 108,431) (World Health Organization, 2026), and short-term exposure continues to drive measurable mortality in Brazilian cities (Requia et al., 2024). When a manager delegates a normative lookup or an episode recommendation to an LLM, an inaccurate answer can propagate into a regulatory brief or an emergency response.

We do not claim the model is least reliable where consequences are greatest, because we did not measure it: disease burden is not among our country covariates, and the one country-level observation we have points the other way, since India carries one of the largest $PM_{2.5}$ mortality burdens in the world and is also the highest-scoring country in our sample. Whether reliability is misaligned with need is answerable by adding an age-standardised attributable-mortality covariate, but not by this design. We interpret the pattern we *did* measure through the coloniality-of-knowledge framework (Mohamed et al., 2020): LLMs are compressed summaries of corpora produced under uneven digital and publication access, and the language penalty is a measurable link from corpus asymmetry to knowledge asymmetry.

5.7. *Limitations and future opportunities*

Each limit below bounds this design, and each names the study that would remove it. We list them in the order we would run them.

Role framing was tested in the user turn, not the system prompt. The manager frame is prefixed to the message rather than installed as a system instruction, where three of the four vendor guides place it — Google's admits either that slot or the start of the user prompt, which is the one we used. The choice follows the population we care about — public servants in the chat interface, who have no system prompt to configure — but our null bounds that user's channel, not a developer's. Whether a system-instruction persona recovers factual accuracy where a typed one does not is a clean two-cell experiment on the same items, and it is the first study we would run.

From panel-scored to gold-anchored. No human expert re-scored responses here. Three properties bound what that costs: where the answer is a register value the verdict is computed by code; where judgement is required the score is the mean of three vendors rather than one; and every reported effect is a between-group contrast, in which a judge's systematic severity cancels. Restricting to objective factual accuracy *strengthens* both headline effects, the opposite of what a permissive judge would produce. A blinded human-expert validation of a stratified sample — roughly 150 responses scored by three raters, which the released pipeline exports directly — would convert an official-source-anchored result into a gold-standard-anchored one.

From a within-country signal to a causal design. Within countries we remove country-level confounding without removing task-by-country confounding. A design that varies country coverage while holding development fixed, or a longitudinal study tracking accuracy as a country's encyclopedic footprint grows, would close that gap.

From a null gradient to an adequately powered test. The tier difference is established while the gradient is not, at $n = 25$. A pre-registered replication at substantially larger n, which the open dataset makes cheap, would settle whether the gradient is genuinely absent or still underpowered.

From three native languages to the full multilingual matrix. The strongest result rests on three native languages, so its *mechanism* is reported descriptively even though the effect is not. The translation-quality objection is settled empirically by the back-translation census reported above; what remains open is breadth, not validity, and scaling to more languages is the highest-value extension this study points to.

From one domain to a cross-domain program. All effects come from one high-stakes applied domain and one corpus family (Wikimedia), and we confine the mechanistic language accordingly. The same task suite and ground-truth-registry method transfer directly to other regulated, source-anchored domains — water quality, food safety, vaccine policy.

Served-model stack as the unit of analysis. We evaluate models *as served* through fixed access stacks, which is what a public manager queries; a residual gap could in principle reflect the stack rather than the weights. A controlled quasi-isolation design with identical open weights across stacks would separate the two, which the per-call provenance we release already enables.

5.8. What this study contributes

The three fixes practitioners reach for first fail together: the local language actively harms, the persona frame does nothing, and the regional model is the single worst performer of fourteen. A benchmark's most useful output is often telling the field what does not work, with effect sizes behind it. The methodological result is that in this literature the ground-truth key, not the sample size, is the binding constraint. And the design separates what 25 countries can demonstrate from what they cannot, reporting the second set as open rather than as supported.

6. Conclusion

Large language models are becoming working infrastructure for air pollution policy, a domain where an inaccurate answer about a binding standard, a measured concentration, or an operational response carries direct public-health stakes in the Global South, where the $PM_{2.5}$ mortality burden is heaviest (GBD 2019 Risk Factors Collaborators, 2020; Requia et al., 2024). We measured geographic and persona-modulated bias in that domain with a 25-country, 14-model, four-language benchmark, scored against ground truth anchored to official primary sources — by code wherever the answer is a register value, and by the mean of a three-vendor judge panel elsewhere.

The remedies an agency can try do not work, and the most intuitive one backfires. Asking in the country's own language *lowers* accuracy by 4.8 percentage points, most sharply where the language is least represented in training corpora; a local-manager persona does not narrow the gap; and the regionally tuned model is the weakest of all fourteen systems. Accuracy also collapses on the recall of a binding value, the task a manager can least afford to have wrong. A Global North/South tier gap persists and is highly structured: on returning the binding national standard the Global South odds of a correct answer are 0.22 of the Global North's, and the gap shrinks as the answer depends less on a country-specific fact, disappearing on the one task with no register value to get right. The monotonic development gradient does not reach significance, so we report a difference between groups that we can demonstrate and a shape across the development range that we cannot. On mechanism, country representation is the operative channel for a gap measured in English, identified within countries rather than between them, and without a claim to causal identification.

For Global South environmental agencies the practical message is cautionary. LLM outputs on binding standards and measured data should be treated as drafts checked against the official gazette, and none of the cheap prompt- or model-level workarounds substitutes for that verification — least of all switching to the local language, which on this evidence makes the answer worse. We interpret the pattern through the coloniality-of-knowledge framework (Mohamed et al., 2020): the languages under-served in training corpora (Joshi et al., 2020) are spoken where air pollution kills the most people, and this study turns that link from a plausible worry into a measured effect.

One methodological result generalises beyond the domain. Rebuilding two task keys from official registers and moving range comparison out of the judge and into code changed the substantive conclusions on the same responses. When an effect in this literature proves fragile, the ground-truth key deserves the audit before the sample does, because a defective key adds *structured* noise that concentrates exactly where official data are hardest to obtain.

Protocol, prompts, official-source ground-truth registries, analysis code, and anonymised response data are released openly, with a one-command reproduction pipeline and a method-audit gate that recomputes every reported number from the raw data. The study also provides a reusable template for applied LLM-equity audits — official-source ground truth, code adjudication wherever the answer is a checkable number, and end-to-end reproducibility — in the high-stakes public-sector domains where these systems are now being deployed.

Data and code availability

The full protocol, prompts, analysis code, ground-truth registries, and the anonymised LLM responses will be released on publication at github.com/Roverlucas/artigo-vies-analise-regional under MIT (code) and CC-BY-4.0 (data, prompts, documentation) licenses. The release includes the per-response records that every reported effect is computed from, so each number in this paper can be recomputed from the raw data rather than taken on trust. The analysis pipeline runs end-to-end via `python code/run_all.py -confirmatory`.

Study protocol

Hypotheses, sampling frame, outcome definition, decision thresholds and multiplicity correction were fixed in a version-controlled analysis plan before confirmatory collection began. That plan was not deposited in a public registry, so readers cannot verify its timing independently of the repository history, and we do not claim pre-registration. Delivered scope also departed substantially from the planned design, which the plan itself made a condition for confirmatory status.

We therefore report the study in two layers. The first states what was specified in advance and what became of it: the three co-primary hypotheses and the secondary hypothesis are reported against their declared thresholds, including the three that failed and the one whose registered direction the data reverse. The second reports what the data suggested and the plan did not anticipate, labelled exploratory throughout, with intervals and without confirmatory language. Deviations from the plan are tabulated in the supplement.

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Author contributions

Following CRediT taxonomy:

- Lucas Rover (UTFPR): Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Visualization, Writing – Original Draft, Project Administration.
- Vitor de Melo Dominski (Faculdade Descomplica): Data Curation, Software, Formal Analysis.
- Prof. Anibal Tavares de Azevedo (Unicamp): Writing – Original Draft, Writing – Review & Editing.

- Dr. Eduardo Tadeu Bacalhau (UFPR): Validation, Supervision, Writing – Review & Editing.
- Dra. Yara de Souza Tadano (UTFPR): Validation, Supervision, Writing – Review & Editing.

Declaration of generative AI in the writing process

Two distinct uses must be separated here, because only one of them is what this declaration is about.

As an object and instrument of the study. Large language models are the subject of this research, and they also serve as one of its measuring instruments: GPT-5-mini, Gemini 2.5 Pro, Claude Sonnet 4.6 and DeepSeek-V3 scored model responses against the ground-truth registry, and Claude Sonnet 4.6 produced the native-language renderings of the prompts. These uses are part of the method, are described in Section 3, and are why the study also reports inter-judge reliability and a back-translation audit of the translated prompts. Wherever the correct answer is a value in an official register, the verdict is computed by code rather than by a model.

In preparing the manuscript. During the preparation of this work the authors used large language models to assist with drafting and editing the text and with implementing analysis code. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No language model is listed as an author, and no model contributed the interpretation of findings or the decisions about what the evidence supports.

Declaration of interests

The authors declare no competing interests.

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Table 1: The five tasks, the prompt as issued, and the decision each stands in for. Prompts are shown for the Brazil instance under the neutral persona; every country receives the same five templates with its own entities substituted (city, agency, regulation).

Task	Prompt as issued (Brazil)	Decision it stands in for
T1 Technical standard	"What is the current national annual mean ambient air quality standard for $PM_{2.5}$ in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences."	Citing the binding limit in a licence condition, an enforcement notice or a technical opinion. The wrong value voids the act.
T2 Local factual datum	"According to the national or subnational environmental agency, what was the annual mean $PM_{2.5}$ concentration in São Paulo, Brazil, in the most recent reported year? Provide value in $\mu g/m^3$, the reference station or network, and the source."	Establishing the baseline a local air quality plan or a public report is written against.
T3 Health-evidence synthesis	"Summarize the epidemiological evidence on mortality attributable to ambient $PM_{2.5}$ exposure in Brazil over the last decade. Include estimated annual deaths, primary causes, and at least one authoritative source."	Justifying why air quality should be prioritised against competing demands on the same budget.
T4 Policy instruments	"List the principal national programs or legal instruments for monitoring and controlling ambient air pollution in Brazil, with the executing agency for each."	Identifying which programme to draw on and which agency holds the mandate.
T5 Applied recommendation	"A metropolitan region of Brazil faces a forecast episode with $PM_{2.5}$ expected to exceed 75 $\mu g/m^3$ within 24 hours. Recommend three short-term policy actions (deployable within 48 hours) to reduce population exposure, each justified by public-health evidence or operational feasibility."	Choosing the operational response to an acute episode, under time pressure.

Table 2: Composite accuracy by model across 25 countries (code-adjudicated where the answer is a register value, panel mean elsewhere; $[0, 1]$). N = scored English-prompt responses.

Model	N	Mean
DeepSeek-V3	500	0.680
GPT-5	500	0.670
Gemini 2.5 Flash	490	0.624
GPT-5-mini	499	0.619
Claude Haiku 4.5	490	0.580
Qwen3 32B	498	0.553
Command R+	500	0.526
Llama 3.3 70B	395	0.507
Llama 4 Scout	500	0.492
Qwen3 14B	500	0.480
Phi-4 14B	500	0.473
GPT-OSS 120B	261	0.472
Llama 3.1 8B	500	0.435
Cabra-Mistral 7B	496	0.347

Table 3: Composite accuracy by task type (25 countries). T2 and T3 are code-adjudicated where the returned value is checkable against the official register; T4 is panel-scored; T1 and T5 retain their original scores.

Task	N	Mean
T5 (applied recommendation)	1,320	0.769
T3 (health-evidence synth.)	1,329	0.548
T4 (policy instruments)	1,310	0.520
T2 (local factual datum)	1,335	0.473
T1 (technical standard)	1,335	0.369

Table 4: H2: native-minus-English accuracy by native language (paired within country and model, replicates averaged within cell, 25-country sample).

Native language (Joshi class)	Countries	Δ (pp)	p	N cells
Spanish (5)	MEX, ARG, PER, COL, CHL	-4.4	$< 10^{-4}$	457
Portuguese (4)	BRA, PRT, AGO	-3.9	2×10^{-4}	311
Hindi (4)	IND	-11.1	5×10^{-4}	71
Overall		-4.8	3×10^{-15}	839

Table 5: Global South versus Global North odds of returning the register value, by task. Binomial mixed model, crossed random intercepts for country and model.

Task	Answer is	GN	GS	OR	95% interval
T1 (technical standard)	a national value	52.2%	29.4%	0.22	$[0.18, 0.27]$
T4 (policy instruments)	national instruments	62.0%	44.7%	0.33	$[0.26, 0.40]$
T2 (local measured datum)	a national value	50.7%	37.8%	0.47	$[0.39, 0.55]$
T3 (health-evidence synth.)	an international estimate	59.1%	52.3%	0.69	$[0.59, 0.80]$
T5 (applied recommendation)	no register value	99.6%	99.7%	3.58	$[0.84, 15.17]$

Table 6: Country coverage × task dependence, response-level mixed model with a fixed effect for country. Positive β = greater coverage, smaller deficit on country-dependent tasks.

Task classification	β	p
T1, T2, T4 versus T3, T5 (primary)	+0.028	7×10^{-9}
T1 versus T5 (purest contrast)	+0.038	2×10^{-9}
Excluding T5 (ceiling at 99.7%)	+0.024	8×10^{-5}
Excluding T4 (most interpretive)	+0.031	1×10^{-8}
Placebo: labels inverted	−0.028	7×10^{-9}

Table 7: Hypotheses, findings, candidate mechanisms, and evidence status. Grouped by what the evidence supports, not by hypothesis number.

Hypothesis	Finding	Mechanism
Supported		
H2 language	-4.8 pp; Hindi -11.1	language corpus
H3 regional model	$\delta = -0.47$, worst of 14	scale over tuning
H1 tier gap	$+5.4$ pp; OR 0.22 on T1	tier
Supported, but not causally identified		
H4 within country	$\beta = +0.028$, $p = 7 \times 10^{-9}$	country coverage
Not supported		
H1 gradient	$\rho = 0.41$, below the 0.55 threshold	development
H4 pre-specified	language-corpus proxy null	language corpus
H6 persona	DiD $+0.8$ pp, $p = 0.29$	—
Descriptive only		
H5 open/closed	$+12.6$ pp for closed	access type

Note. H2's mechanism rests on three native languages and is descriptive. H1's tier gap is concentrated in tasks whose answer depends on a national fact. H4 within country removes country-level confounding by construction, which is not causal identification.